

Contest Math 8: Intro to Geometry (27079)

Connor Gordon

Sunday

Feb 16, 2025 - Nov 9, 2025

12:45 - 2:45 PM ET

Homework

Meeting: 1 2 3 4 **5** 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Homework: Meeting 5

You have completed **9** of **15** challenge problems.
 Due **Mar 23**.

Readings

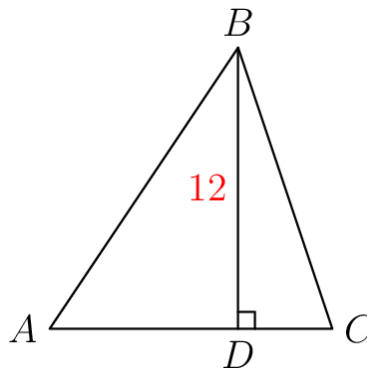
Meeting 5: Chapter 4**Meeting 6:** Sections 5.1-5.4

Challenge Problems

Problem 1 – Correct! – Score: 7 / 7 (28329)

Problem:[Report Error](#)

Triangle ABC has area 16. Find length AC .



(Based on problem 4.9 in the textbook.)

Solution:

We can compute the area of ABC as $\frac{1}{2} \cdot BD \cdot AC$. Since $BD = 12$, we have

$$\frac{1}{2} \cdot 12 \cdot AC = 16,$$

$$\text{so } AC = \frac{16}{6} = \boxed{\frac{8}{3}}.$$

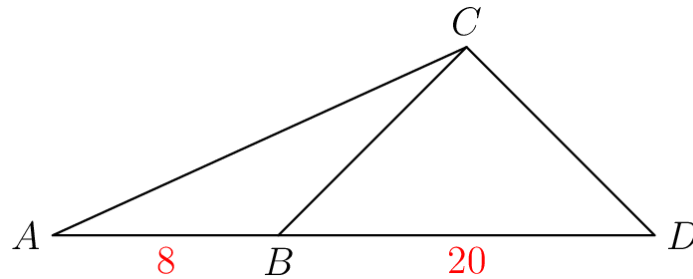
Your Response(s):

8/3

Problem 2 – Correct! – Score: 7 / 7 (28330)

Problem:[Report Error](#)

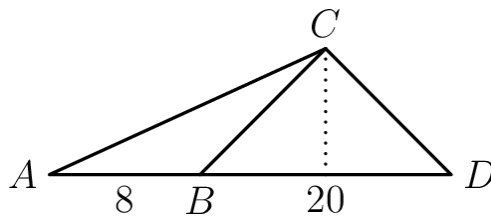
Given that $[ABC] = 10$, find $[BCD]$.



(Based on problem 4.10 in the textbook.)

Solution:

We drop an altitude from C to $\overline{AD}$.



Let the length of this altitude be h . Since the area of ABC is 10, we know $\frac{1}{2} \cdot 8 \cdot h = 10$, or $h = \frac{20}{8} = \frac{5}{2}$. Since the dotted line is also a height of triangle BCD , we see that BCD has area $\frac{1}{2} \cdot 20 \cdot \frac{5}{2} = \boxed{25}$.

We can also use the fact if two triangles have the same height, then the ratio of their areas is equal to the ratio of their bases, so

$$\frac{[BCD]}{[ABC]} = \frac{BD}{AB}.$$

$$\text{Thus, } [BCD] = [ABC] \cdot \frac{BD}{AB} = 10 \cdot \frac{20}{8} = \boxed{25}.$$

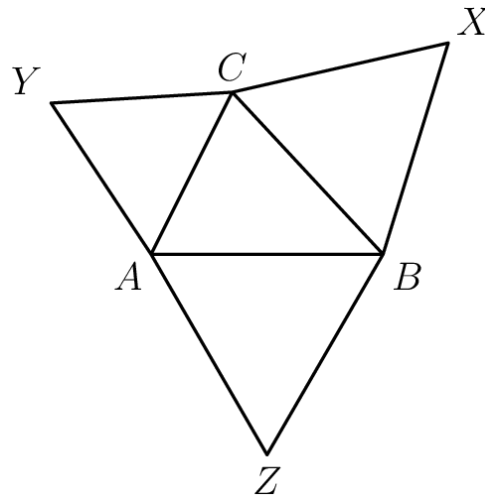
Your Response(s):

☺ 25

Problem 3 – Correct! – Score: 7 / 7 (10390)

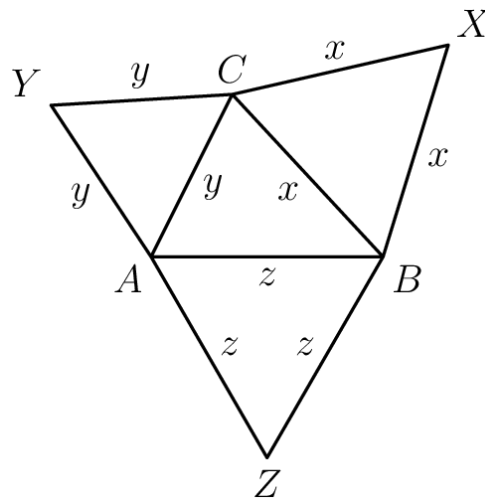
Problem:[Report Error](#)

Let ABC be any triangle. Equilateral triangles BCX , ACY , and BAZ are constructed such that none of these triangles overlaps triangle ABC . The perimeter of hexagon $AZBXC Y$ is 60. What is the perimeter of triangle ABC ?

**Solution:**

Label the lengths of line segments as shown in the diagram. We know the perimeter of hexagon $AZBXC Y$ is $x + x + y + y + z + z$, or $2x + 2y + 2z$. Since that perimeter is equal to 60, we have

$x + y + z = 60/2 = 30$. Since $x + y + z$ is also the perimeter of $\triangle ABC$, our answer is 30.

**Your Response(s):**

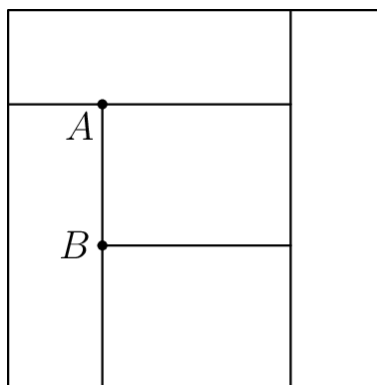
😊 30

Problem 4 – Correct! – Score: 7 / 7 (39005)

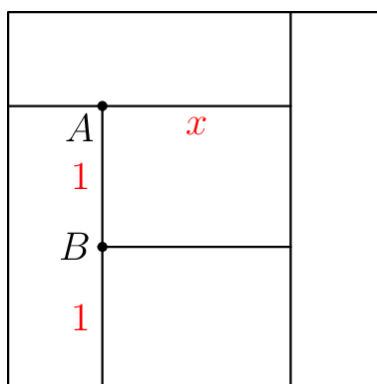
Problem:

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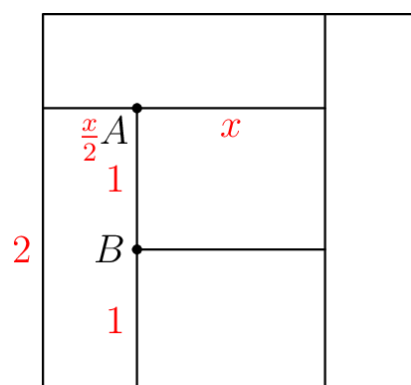
A square is divided into five rectangles, and all the rectangles have the same area. Given that $AB = 1$, find the side length of the square.

**Solution:**

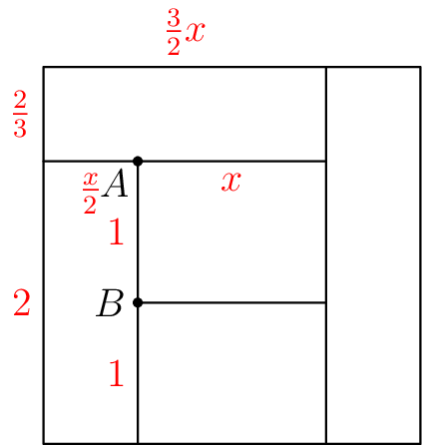
Let x be the width of the center rectangle. Then the area of the center rectangle is x , so all the rectangles have an area of x . In turn, this implies that the height of the lower rectangle is also 1.



Then the left rectangle has a height of 2, so its width is $\frac{x}{2}$.



Then the top rectangle has a width of $\frac{x}{2} + x = \frac{3}{2}x$, so its height is $\frac{2}{3}$.

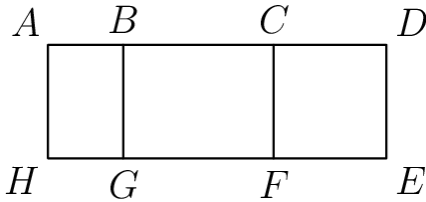


Thus, the side length of the square is $\frac{2}{3} + 2 = \boxed{\frac{8}{3}}$.

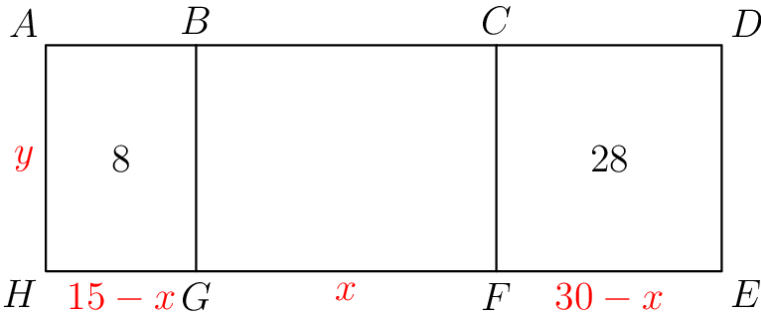
Your Response(s):
😊 8/3

Problem 5 – Correct! – Score: 7 / 7 (39004)

Problem: Rectangle $ADEH$ is divided into rectangles $ABGH$, $BCFG$, and $CDEF$. If $AC = 15$, $GE = 30$, $[ABGH] = 8$, and $[CDEF] = 28$, find $[BCFG]$. Report Error



Solution: Let $x = BC = GF$. Then $AB = HG = 15 - x$ and $CD = FE = 30 - x$. Let $y = AH$.



Then $y(15 - x) = 8$ and $y(30 - x) = 28$, or

$$15y - xy = 8,$$

$$30y - xy = 28.$$

Subtracting these equations, we get $15y = 20$, so $y = \frac{20}{15} = \frac{4}{3}$. Substituting into the equation $y(15 - x) = 8$, we get $\frac{4}{3}(15 - x) = 8$. Then $15 - x = 6$, so $x = 9$. Therefore, $[BCFG] = xy = 9 \cdot \frac{4}{3} = \boxed{12}$.

Your Response(s):

😊 12

Problem 6 – Correct! – Score: 7 / 7 (10398)

Problem:

[Report Error](#)

A rectangle is divided into four small rectangles as shown. Three of the small rectangles are labelled with their **perimeter**. What is the perimeter of the fourth rectangle?

10	?
6	25

Solution:

Label lengths a , b , c , and d , as shown below.

	c	d
a	10	?
b	6	25

Then

$$2a + 2c = 10,$$

$$2b + 2c = 6,$$

$$2b + 2d = 25.$$

Adding the equations $2a + 2c = 10$ and $2b + 2d = 25$, we get $2a + 2b + 2c + 2d = 35$. Subtracting the equation $2b + 2c = 6$, we get $2a + 2d = \boxed{29}$, which is the perimeter of the fourth rectangle.

Your Response(s):

😊 29

Problem 7 – Correct! – Score: 7 / 7 (10391)

Problem: A rectangle is divided into four small rectangles as shown. Three of the small rectangles are labelled with their **area**. What is the area of the fourth rectangle?

12	?
8	27

Solution:
Label lengths a, b, c , and d , as shown below.

	c	d
a	12	?
b	8	27

Then

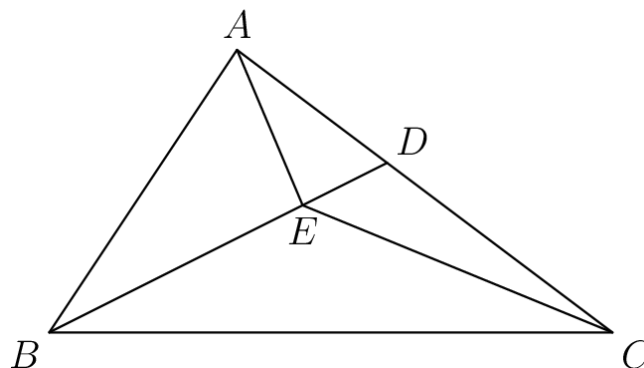
$$\begin{aligned}ac &= 12, \\bc &= 8, \\bd &= 27.\end{aligned}$$

Multiplying the equations $ac = 12$ and $bd = 27$, we get $abcd = 324$. Dividing the equation $bc = 8$, we get $ad = \frac{324}{8} = \frac{81}{2}$, which is the area of the fourth rectangle.

Your Response(s):
😊 40.5

Problem 8 – Correct! – Score: 7 / 7 (39008)

Problem: In triangle ABC , $[ADE] = 4$, $[ABE] = 14$, and $[CDE] = 6$. Compute $[BCE]$.

**Solution:**

Triangles ABE and ADE have the same height with respect to line BD , so

$$\frac{BE}{DE} = \frac{[ABE]}{[ADE]} = \frac{14}{4} = \frac{7}{2}.$$

Triangles BCE and CDE have the same height with respect to line BD , so

$$\frac{BE}{DE} = \frac{[BCE]}{[CDE]} = \frac{[BCE]}{6}.$$

Hence, $[BCE] = 6 \cdot \frac{7}{2} = \boxed{21}$.

Your Response(s):

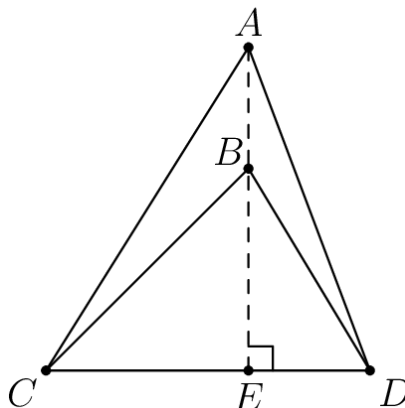
☺ 21

Problem 9 – Correct! – Score: 5 / 7 (10385)

Problem:

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In the diagram, $BE = 7$, $AB = 6$, and $[CBDA] = 15$. What is $[CDA]$?

**Solution:**

Since triangle ABC and triangle BEC have the same height (CE), we can say that

$$\frac{[ABC]}{[BEC]} = \frac{AB}{BE} = \frac{6}{7}.$$

Similarly, $\frac{[ABD]}{[BED]} = \frac{AB}{BE} = \frac{6}{7}$.

Putting these facts together, we can see that

$$\begin{aligned} [CBDA] &= [ABC] + [ABD] \\ &= \frac{6}{7}[BEC] + \frac{6}{7}[BED] \\ &= \frac{6}{7}([BEC] + [BED]) \\ &= \frac{6}{7}[BCD]. \end{aligned}$$

The fact that $[CBDA] = 15$ now tells us that $[BCD] = \frac{7}{6} \cdot 15 = \frac{35}{2}$.

Thus, $[CDA] = [BCD] + [CBDA] = \frac{35}{2} + 15 = \boxed{\frac{65}{2}}$.

Your Response(s):

☹ 22.5

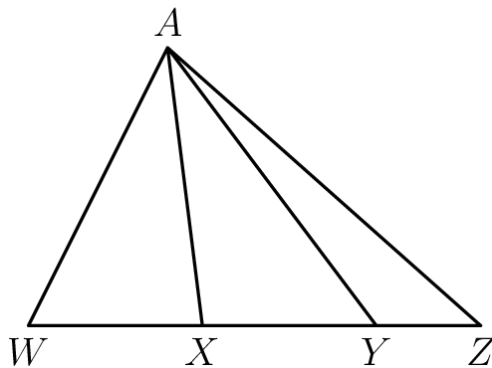
😊 65/2

Problem 10 (10389)

Problem:

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In the diagram, $WY = 12$, $XZ = 10$, $[AWX] = 15$, and $[AYZ] = 8$. Find $[AXY]$.



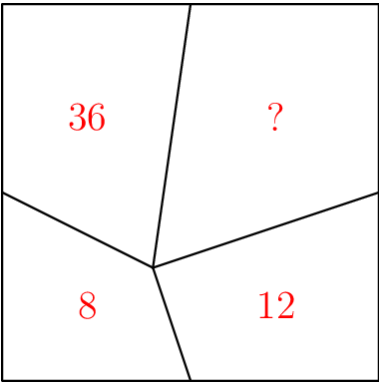
Your Answer

Problem 11 (39006)

Problem:

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A point in a square is connected to the midpoints of its sides, dividing the square into four regions. The areas of three of the regions are labelled. Find the area of the fourth region.

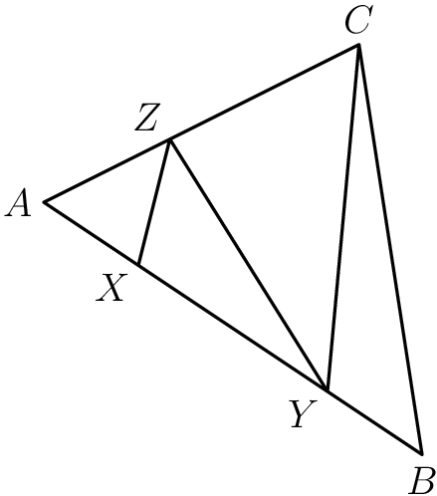


Your Answer

Problem 12 (10417)

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Problem:
In the diagram, $XY = 2 \cdot AX = 2 \cdot YB$ and $\frac{AZ}{ZC} = \frac{2}{5}$. Find $\frac{[YBC]}{[ZYC]}$.



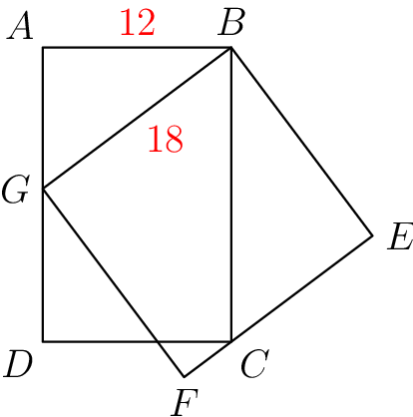
Your Answer

[Tips](#)

Problem 13 (39000)

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Problem:
In the diagram, $ABCD$ is a rectangle, and $BEFG$ is a square. Find AD .



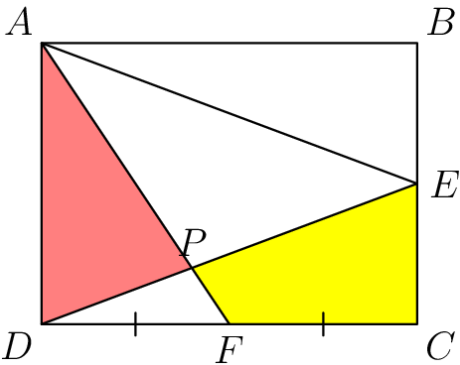
Your Answer

Problem 14 (39009)

Problem:

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In rectangle $ABCD$, $\frac{[AEP]}{[DFP]} = 5$. Find the ratio of the area of the red region to the area of the yellow region. Enter your answer as a fraction.



Your Answer

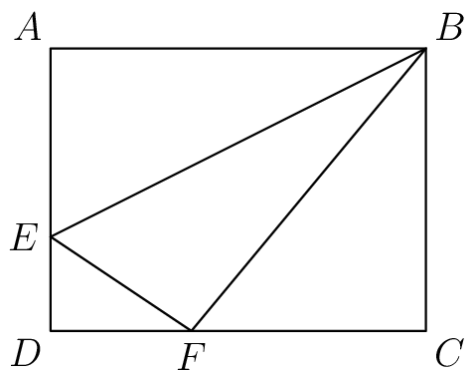
[Tips](#)

Problem 15 (39007)

Problem:

[Report Error](#)

In rectangle $ABCD$, $[ABE] = [BEF] = [BFC]$. Compute $\frac{AE}{ED}$.

**Beast Problem:** 🐾

This is a **Beast** problem. Answering this problem can only increase your score. You will not lose points for skipping this problem or answering incorrectly.

[Tips](#)

Book Problems (39027)

In addition to the homework listed here each week, you also need to complete several additional homework problems from your textbooks. This week's book problems are:

- *Introduction to Geometry*: 4.23, 4.24, 4.32, 4.38
- *Art of Problem Solving, Vol. 1*: Problem 265

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